

API RESTful Deployment and Testing Lab

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Implementation Summary

Task 8.1 and Task 8.2 are implemented as a no-key REST API system using Express.js. Weather data comes from Open-Meteo, and country headlines come from Google News RSS converted into structured JSON.

- `GET /api/weather/:city` returns city name, current temperature, weather condition, and humidity level.
- `GET /api/news/:countryCode?limit=6` returns 5-10 headlines with title, source, URL, and publication date.
- Invalid inputs and provider failures return structured error JSON.
- Jest unit, integration, and system tests are included.

Task Screenshots

Task 8.1 Weather UI

Full Stack Programming Lab 14



REST API testing console

Weather
Open-Meteo

News
Google RSS

Tests
Jest verified

WEATHER

NEWS

JSON

8.1
Weather

8.2
News

Task 8.1

Weather Forecast API



City name

 Run

GET http://127.0.0.1:5014/api/weather/Karachi

Weather output

Current city snapshot



Current temperature

Humidity level

Weather condition

32.6 C

66%

Thunderstorm

City name

Observed

Coordinates

Provider

Karachi, Pakistan

Jun 02, 04:15 PM

24.8608, 67.0104

Open-Meteo

Task 8.1 API JSON

http://127.0.0.1:5014/api/weather/Karachi



```
{
  "success": true,
  "task": "8.1",
  "data": {
    "city": "Karachi",
    "country": "Pakistan",
    "countryCode": "PK",
    "coordinates": {
      "latitude": 24.8608,
      "longitude": 67.0104
    },
    "currentTemperature": 32.6,
    "temperatureUnit": "Celsius",
    "temperatureCelsius": 32.6,
    "weatherCondition": "Thunderstorm",
    "condition": "Thunderstorm",
    "humidityLevel": 66,
    "humidityUnit": "%",
    "humidity": 66,
    "observedAt": "2026-06-02T16:15",
    "fetchedAt": "2026-06-02T11:21:22.119Z",
  }
}
```

Open API ↗

Task 8.1 Weather API JSON

Pretty-print ☐

```
{
  "success": true,
  "task": "8.1",
  "data": {
    "city": "Karachi",
    "country": "Pakistan",
    "countryCode": "PK",
    "coordinates": {
      "latitude": 24.8608,
      "longitude": 67.0104
    },
    "currentTemperature": 32.8,
    "temperatureUnit": "Celsius",
    "temperatureCelsius": 32.8,
    "weatherCondition": "Thunderstorm",
    "condition": "Thunderstorm",
    "humidityLevel": 65,
    "humidityUnit": "%",
    "humidity": 65,
    "observedAt": "2026-06-02T16:00",
    "fetchedAt": "2026-06-02T11:08:25.520Z",
    "provider": "Open-Meteo"
  }
}
```

Task 8.2 News UI

News output

Latest country headlines

The Washington Post

Jun 02, 03:27 PM

Iran says it is breaking off ceasefire talks over Israeli attacks on Lebanon

Open source 

Business Recorder

Jun 02, 12:13 PM

At UNSC, Pakistan demands urgent global action to halt Israel's Lebanon incursion

Open source 

Dawn

Jun 02, 07:24 AM

BUDGET 2026-27: IMF-mandated curbs squeeze development spending

Open source 

The Guardian

Jun 02, 02:46 PM

Prepare for imminent return of El Niño, UN warns

Open source 

Reuters

Jun 02, 03:33 PM

Kremlin talks of new 'paradigm' in Ukraine war, accusing Kyiv of 'acts of terror'

Open source 

Al Jazeera

Jun 02, 11:58 AM

Iran war day 95: Trump pushes Lebanon truce after Tehran vows to end talks

Open source 

```
Task 8.2 API JSON
http://127.0.0.1:5014/api/news/PK?limit=6

{
  "success": true,
  "task": "8.2",
  "data": {
    "countryCode": "PK",
    "count": 6,
    "articles": [
      {
        "newsTitle": "Iran says it is breaking off ceasefire talks over Israeli attacks on Lebanon",
        "title": "Iran says it is breaking off ceasefire talks over Israeli attacks on Lebanon",
        "sourceName": "The Washington Post",
        "newsUrl": "https://news.google.com/rss/articles/CBMioAFBV95cUXPmq0zc1ZESDFjVJ3cUZiOUhhQVFUjFjdIV34bFoQkVb3Z3dhFVE11SGNGaUdsQTQxN2U3OwUe1QwbmdKOH12VUZ5bjVWmNMSTDB5dmcF12VWk1mUnV
mVndoQ85hV1FVzNYOEFmeF1YOX3Fd95XmNvKtUFERiprWmZYOT1NaTd1MmctU1lpNE910Xk1M8RRNzFTVXYT?oc=5",
        "url": "https://news.google.com/rss/articles/CBMioAFBV95cUXPmq0zc1ZESDFjVJ3cUZiOUhhQVFUjFjdIV34bFoQkVb3Z3dhFVE11SGNGaUdsQTQxN2U3OwUe1QwbmdKOH12VUZ5bjVWmNMSTDB5dmcF12VWk1mUnV
mVndoQ85hV1FVzNYOEFmeF1YOX3Fd95XmNvKtUFERiprWmZYOT1NaTd1MmctU1lpNE910Xk1M8RRNzFTVXYT?oc=5",
        "publicationDate": "2026-06-02T10:27:23.000Z",
        "publishedAt": "2026-06-02T10:27:23.000Z"
      },
      {

```

Task 8.2 News API JSON

```
{
  "success": true,
  "task": "8.2.",
  "data": {
    "countryCode": "PK",
    "count": 6,
    "articles": [
      {
        "newsTitle": "Iran says it is breaking off ceasefire talks over Israeli attacks on Lebanon",
        "title": "Iran says it is breaking off ceasefire talks over Israeli attacks on Lebanon",
        "sourceName": "The Washington Post",
        "newsUrl": "https://www.news.google.com/rss/articles/CBMIA0AFB9V95CkPmQw0c1ZESDFJvZC2UiohHQVFJvD1V3V4b6FqKjVb3Z3dHbFV1LSGNgaGdsQTXQNZ30WtueIqubmcdKH1ZVJ5b3jVwMNSTB5dTmcF1ZK1uNmVnVmDQ085Vt1Fv2NoYfEeF1VOX3FD05XmNkVFtERipmHYZ0T1Aid1MctUit1PNE91Kh1MRR9Z1FVXYT?oc=5",
        "url": "https://www.news.google.com/rss/articles/CBMIA0AFB9V95CkPmQw0c1ZESDFJvZC2UiohHQVFJvD1V3V4b6FqKjVb3Z3dHbFV1LSGNgaGdsQTXQNZ30WtueIqubmcdKH1ZVJ5b3jVwMNSTB5dTmcF1ZK1uNmVnVmDQ085Vt1Fv2NoYfEeF1VOX3FD05XmNkVFtERipmHYZ0T1Aid1MctUit1PNE91Kh1MRR9Z1FVXYT?oc=5",
        "publishedDate": "2026-06-02T18:27:23.000Z",
        "newsTitle": "IMF-mandated curbs squeeze developer spending",
        "title": "IMF-mandated curbs squeeze developer spending",
        "sourceName": "The Washington Post",
        "newsUrl": "https://www.news.google.com/rss/articles/CBM1SEF5V31XtE1YK1aUvFc3j1VwHvMndrZRFVFC10SF9RQ1M1WjhJd1N1UoyCk5Uxp11LSR0vZcXKXUH8ReGtvTHTQ0qvbD1BTKFVX31xT80TE1nZGc5pR1LNKbXuaBvZk10R6Z0z01n1T93R1pRHJ3Y10XzhkVvHhGks15U1NmXusV1jPa6g2K132Uk1Ek1F52TVYUQ?oc=5",
        "url": "https://www.news.google.com/rss/articles/CBM1SEF5V31XtE1YK1aUvFc3j1VwHvMndrZRFVFC10SF9RQ1M1WjhJd1N1UoyCk5Uxp11LSR0vZcXKXUH8ReGtvTHTQ0qvbD1BTKFVX31xT80TE1nZGc5pR1LNKbXuaBvZk10R6Z0z01n1T93R1pRHJ3Y10XzhkVvHhGks15U1NmXusV1jPa6g2K132Uk1Ek1F52TVYUQ?oc=5",
        "publishedDate": "2026-06-02T18:24:36.000Z",
        "newsTitle": "Trump says Hezbollah and Israel have agreed to 'stop all shooting'",
        "title": "Trump says Hezbollah and Israel have agreed to 'stop all shooting'",
        "sourceName": "The Guardian",
        "newsUrl": "https://www.news.google.com/rss/articles/CBM1FVX31xT80TE1nZGc5pR1LNKbXuaBvZk10R6Z0z01n1T93R1pRHJ3Y10XzhkVvHhGks15U1NmXusV1jPa6g2K132Uk1Ek1F52TVYUQ?oc=5",
        "url": "https://www.news.google.com/rss/articles/CBM1FVX31xT80TE1nZGc5pR1LNKbXuaBvZk10R6Z0z01n1T93R1pRHJ3Y10XzhkVvHhGks15U1NmXusV1jPa6g2K132Uk1Ek1F52TVYUQ?oc=5",
        "publishedDate": "2026-06-01T18:56:00.000Z",
        "newsTitle": "Pakistan demands urgent global action to halt Israel's Lebanon incursion",
        "title": "Pakistan demands urgent global action to halt Israel's Lebanon incursion",
        "sourceName": "The Business Recorder",
        "newsUrl": "https://www.news.google.com/rss/articles/CBM1FVX31xT80TE1nZGc5pR1LNKbXuaBvZk10R6Z0z01n1T93R1pRHJ3Y10XzhkVvHhGks15U1NmXusV1jPa6g2K132Uk1Ek1F52TVYUQ?oc=5",
        "url": "https://www.news.google.com/rss/articles/CBM1FVX31xT80TE1nZGc5pR1LNKbXuaBvZk10R6Z0z01n1T93R1pRHJ3Y10XzhkVvHhGks15U1NmXusV1jPa6g2K132Uk1Ek1F52TVYUQ?oc=5",
        "publishedDate": "2026-06-02T07:13:16.000Z",
        "newsTitle": "Prepare for imminent return of El Ni\u00f1o, UN warns",
        "title": "Prepare for imminent return of El Ni\u00f1o, UN warns",
        "sourceName": "The Guardian",
        "newsUrl": "https://www.news.google.com/rss/articles/CBM1A0FBV95CkPmQw0c1ZESDFJvZC2UiohHQVFJvD1V3V4b6FqKjVb3Z3dHbFV1LSGNgaGdsQTXQNZ30WtueIqubmcdKH1ZVJ5b3jVwMNSTB5dTmcF1ZK1uNmVnVmDQ085Vt1Fv2NoYfEeF1VOX3FD05XmNkVFtERipmHYZ0T1Aid1MctUit1PNE91Kh1MRR9Z1FVXYT?oc=5",
        "url": "https://www.news.google.com/rss/articles/CBM1A0FBV95CkPmQw0c1ZESDFJvZC2UiohHQVFJvD1V3V4b6FqKjVb3Z3dHbFV1LSGNgaGdsQTXQNZ30WtueIqubmcdKH1ZVJ5b3jVwMNSTB5dTmcF1ZK1uNmVnVmDQ085Vt1Fv2NoYfEeF1VOX3FD05XmNkVFtERipmHYZ0T1Aid1MctUit1PNE91Kh1MRR9Z1FVXYT?oc=5",
        "publishedDate": "2026-06-02T07:13:16.000Z",
        "newsTitle": "After an Atonizing Week of Threats, Kyiv Is Finally Bombarded by Russia",
        "title": "After an Atonizing Week of Threats, Kyiv Is Finally Bombarded by Russia",
        "sourceName": "The New York Times",
        "newsUrl": "https://www.news.google.com/rss/articles/CBM1FVX31xT80TE1nZGc5pR1LNKbXuaBvZk10R6Z0z01n1T93R1pRHJ3Y10XzhkVvHhGks15U1NmXusV1jPa6g2K132Uk1Ek1F52TVYUQ?oc=5",
        "url": "https://www.news.google.com/rss/articles/CBM1FVX31xT80TE1nZGc5pR1LNKbXuaBvZk10R6Z0z01n1T93R1pRHJ3Y10XzhkVvHhGks15U1NmXusV1jPa6g2K132Uk1Ek1F52TVYUQ?oc=5",
        "publishedDate": "2026-06-02T09:45:00.000Z",
        "newsTitle": "Fetched at: 2026-06-02T11:08:52.204Z",
        "provider": "Google News RSS"
      }
    ]
  }
}
```

Testing Evidence

Jest All Test Suites

Jest Testing - All Lab 14 Suites

=====

```
> lab-14-no-key-api-backend@1.0.0 test
> jest --runInBand --verbose
```

```
Test Suites: 5 passed, 5 total
Tests:      16 passed, 16 total
Snapshots:  0 total
Time:       5.155 s, estimated 6 s
Ran all test suites.
```

Unit Tests

Unit Testing - Task 8.1 and Task 8.2

=====

```
> lab-14-no-key-api-backend@1.0.0 test:unit
> jest tests/unit --runInBand --verbose
```

```
Test Suites: 2 passed, 2 total
Tests:       7 passed, 7 total
Snapshots:   0 total
Time:        1.926 s
Ran all test suites matching tests/unit.
```

Integration Tests

Integration Testing - API Endpoints

=====

```
> lab-14-no-key-api-backend@1.0.0 test:integration
> jest tests/integration --runInBand --verbose
```

```
Test Suites: 2 passed, 2 total
Tests:       7 passed, 7 total
Snapshots:   0 total
Time:        3.763 s, estimated 5 s
Ran all test suites matching tests/integration.
```

System Tests

System Testing - Running HTTP Server

=====

```
> lab-14-no-key-api-backend@1.0.0 test:system
> jest tests/system --runInBand --verbose
```

```
Test Suites: 1 passed, 1 total
Tests:       2 passed, 2 total
Snapshots:   0 total
Time:        2.229 s
Ran all test suites matching tests/system.
```

Browser Testing

Full Stack Programming Lab 14

REST API testing console

Weather Open-Meteo News Google RSS Tests Jest verified

WEATHER NEWS JSON

8.1 Weather 8.2 News

Task 8.1 Weather Forecast API

City name Karachi Q Run

GET http://127.0.0.1:5014/api/weather/Karachi

Weather output Current city snapshot

Current temperature 32.6 C Humidity level 66% Weather condition Thunderstorm

City name Karachi, Pakistan Observed Jun 02, 04:15 PM
Coordinates 24.8608, 67.0104 Provider Open-Meteo

Task 8.1 API JSON
http://127.0.0.1:5014/api/weather/Karachi

```
{
  "success": true,
  "task": "8.1",
  "data": {
    "city": "Karachi",
    "country": "Pakistan",
    "countryCode": "PK"
  }
}
```

Frontend Build

Frontend Build Verification

```
> lab-14-no-key-api-frontend@0.1.0 build
> next build

▲ Next.js 16.2.7 (Turbopack)

Creating an optimized production build ...
✓ Compiled successfully in 7.0s
Running TypeScript ...
Finished TypeScript in 6.4s ...
Collecting page data using 4 workers ...
Generating static pages using 4 workers (0/3) ...
✓ Generating static pages using 4 workers (3/3) in 1123ms
Finalizing page optimization ...

Route (app)
├─ /
└─ /_not-found

○ (Static) prerendered as static content
```

GitHub Repository URL

`https://github.com/abdullahfawadd/Full-stack-programming-lab`